

HTPC SYSTEM REPORT – KDE Neon HTPC (Plasma 6.5.3, Wayland)

===== 1. System Architecture
Overview ----- Operating System / Desktop • KDE Neon (Ubuntu 24.04 base) • Plasma 6.5.3 running on Wayland • HDR requires Wayland; nut.js does not work under Wayland, so ydotool was required. Hardware Signal Chain Intel iGPU (HDMI) → Denon AVR-X2300W → TCL 65R635 Roku TV Playback Architecture • Kodi (ALSA passthrough, HDR) • PipeWire (system audio, Plexamp) • Input backend: ydotool • Web remote: Next.js app + Caddy reverse proxy
----- 2. Display Configuration & EDID Override
----- Goals • 4K60 HDR (scale 2.0) • 1440p120

HDR/SDR (scale 1.5) • Switching controlled from HTPC Remote • Mode must survive kernel updates
EDID Override Details • Custom EDID filename: /lib/firmware/edid/denon-1440p.edid GRUB
Configuration /etc/default/grub:
GRUB_CMDLINE_LINUX="drm.edid_firmware=HDMI-A-1:edid/denon-1440p.edid" Then: sudo
update-grub Persistence Reason • /etc/default/grub is never replaced • /lib/firmware persists across
kernels • initramfs auto-includes reference • No per-kernel copying needed KScreen Storage
~/config/kscreen/ ----- 3. Display Script: htpc-res
----- Location: ~/bin/htpc-res Purpose: • Switch between 4K60 / 1440p120 (HDR/SDR)
• Set scaling (2.0 for 4K, 1.5 for 1440p) • Called by HTPC Remote Notes: • kscreen-

doctor includes ANSI color codes • Must use grep -A60 and robust parsing
----- 4. Audio System ----- Kodi Passthrough
~/config/environment.d/kodi.conf: KODI_AE_SINK=ALSA PipeWire / Plexamp Installing this package
enabled 5.1 output: pipewire-audio-client-libraries Audio Paths Kodi → ALSA → HDMI → Denon
Everything else → PipeWire → Denon/TV ----- 5. Kodi with
PKC ----- Why Kodi: • HDR support • Bitstream HD audio • PKC integrates Plex library
Behavior Notes: • Passthrough only appears if AVR is powered on at startup
----- 6. HTPC Remote App (Next.js) -----

Project Location: ~/WebstormProjects/htpc-remote/ Vended nut.js:
~/WebstormProjects/htpc-remote/vendor/@nut-tree/ Scripts in ~/bin: • htpc-res • htpc-key • htpc-audio •
htpc-launch ----- 7. Systemd User Units -----
Located at: ~/config/systemd/user/ Units: • htpc-remote.service • caddy.service • (optional)
ydotool.service ----- 8. ydotool Input Backend
----- Reasons: • nut.js broken under Wayland • HDR required Wayland • ydotool replaces
nut.js functionality Build Dependencies: • build-essential • cmake • pkg-config • libevdev-dev •
libudev-dev • scdoc Installed Binaries: • /usr/local/bin/ydotool • /usr/local/bin/ydotool

----- 9. System-Level Notes / Gotchas
----- • kscreen-doctor output uses ANSI escapes → parsing unreliable without
cleaning. • systemctl --user cannot be used under sudo. • Kodi passthrough depends on live AVR
connection. • pipewire-audio-client-libraries required for Plexamp 5.1. • Vended nut.js must stay
permanently. ----- Appendix A – Manifest -----
Scripts (~/bin/) • htpc-res • htpc-key • htpc-audio • htpc-launch Systemd User Units
(~/config/systemd/user/) • htpc-remote.service • caddy.service • ydotool.service (if present) EDID /
Display • /lib/firmware/edid/denon-1440p.edid • /etc/default/grub • /boot/grub/grub.cfg

(auto-generated) • ~/config/kscreen/ Audio Config • ~/config/environment.d/kodi.conf • Package:
pipewire-audio-client-libraries HTPC Remote App • ~/WebstormProjects/htpc-remote/ •
~/WebstormProjects/htpc-remote/vendor/@nut-tree/ ydotool • /usr/local/bin/ydotool •
/usr/local/bin/ydotool APT Packages • edid-decode • i2c-tools • pipewire • pipewire-pulse •
pipewire-audio-client-libraries • alsa-utils • libavcodec-extra • kodi • nodejs • npm • build-essential •
cmake • pkg-config • libevdev-dev • libudev-dev • scdoc • caddy END OF REPORT

